OfflineAir

Offline Flight Tracker

Software Requirements Specification

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Updated: Post-flight Sync & Comparison Feature

1. Introduction

1.1 Purpose

This SRS defines the functional and non-functional requirements for OfflineAir, a mobile application that tracks flights offline using dead reckoning, and — once back online — downloads the actual ADS-B flight track to compare against the estimated one.

1.2 Scope

OfflineAir operates across three phases:

- Pre-flight: Download route, waypoints, speed/altitude profiles, and SIGMETs before boarding
- In-flight: Track estimated position offline using dead reckoning. Record every estimated position as a timestamped track
- Post-flight: Once connected, fetch the actual ADS-B track from OpenSky. Compare it against the estimated track and generate an accuracy report

The comparison view shows both tracks overlaid on a map, per-point deviation distances, phase-level accuracy, and an overall accuracy score.

1.3 Definitions

Term	Definition
Dead Reckoning	Estimating position using known starting point, elapsed time, speed, and heading
ADS-B	Automatic Dependent Surveillance-Broadcast — real-time position broadcast by aircraft, recorded by OpenSky
Estimated Track	The sequence of timestamped positions generated by the dead reckoning engine during the flight
Actual Track	The ground-truth ADS-B position data retrieved from OpenSky after the flight
Deviation	Haversine distance (in km) between an estimated point and the nearest actual point at the same timestamp
RMSE	Root Mean Square Error — statistical measure of average deviation across the full track
Accuracy Score	A 0–100 score derived from RMSE, max deviation, and phase alignment
SIGMET	Significant Meteorological Information — pre-downloaded advisory for severe weather
Phase	One of: Climb, Cruise, or Descent — identified by altitude curve shape
PostFlightSync	The module that fetches and normalizes actual ADS-B data after landing

FlightComparator	The module that aligns, diffs, and scores estimated vs. actual tracks

1.4 References

- AviationAPI: https://aviationapi.com
- OpenSky Network REST API: https://opensky-network.org/apidoc
- ADS-B Exchange: https://adsbexchange.com
- Aviation Weather Center: https://aviationweather.gov
- OurAirports Dataset: https://ourairports.com/data/

2. Overall Description

2.1 Three-Phase Product Model

Phase 1: Pre-flight (Online)	Phase 2: In-flight (Offline)	Phase 3: Post-flight (Online)
• Download route, waypoints, speed profile, SIGMETs • Cache offline map tiles • Confirm flight details	• Dead reckon position every 30s • Record estimated track to SQLite • Send landmark and turbulence alerts • Manual position nudge	• Fetch actual ADS-B track from OpenSky • Normalize and align with estimated track • Generate deviation map and accuracy report

2.2 Constraints

- No internet available in-flight — dead reckoning must function with zero network calls
- OpenSky historical data has ~1 hour delay after landing — post-flight sync should retry if data not yet available
- ADS-B coverage is not 100% global — oceanic routes may have sparse actual track data
- Track alignment requires timestamp matching — clocks must be consistent between device and ADS-B source
- Device GPS unavailable in airplane mode — all in-flight positions are estimated only

2.3 Assumptions

- OpenSky Network retains ADS-B tracks for at least 7 days after the flight
- The user will initiate post-flight sync within 7 days of landing
- Sufficient ADS-B receivers existed along the route to produce a usable actual track
- The filed route reflects the path actually flown with only minor deviations

3. Functional Requirements

3.1 Pre-flight Data Download

ID	Description	Priority	Source	Status
FR-01	User can search for a flight by flight number	High	User	Planned
FR-02	App fetches filed route and waypoints from AviationAPI	High	System	Planned
FR-03	App fetches historical speed/altitude profile from OpenSky	High	System	Planned
FR-04	App fetches active SIGMETs along the route	High	System	Planned
FR-05	All pre-flight data is persisted to local SQLite	High	System	Planned
FR-06	Offline map tiles are pre-cached for the flight corridor	High	System	Planned

3.2 In-flight Dead Reckoning & Track Recording

ID	Description	Priority	Source	Status
FR-07	User can start tracking by entering actual takeoff time	High	User	Planned
FR-08	Dead reckoning engine updates position estimate every 30 seconds	High	System	Planned
FR-09	Every position estimate is saved as a timestamped EstimatedTrackPoint in SQLite	High	System	Planned
FR-10	Altitude and speed are estimated and recorded alongside each position	High	System	Planned
FR-11	ETA is recalculated dynamically on every position update	High	System	Planned
FR-12	User can manually nudge estimated position; nudge is recorded in the track	Medium	User	Planned
FR-13	App detects likely landing when altitude curve reaches destination phase end	Medium	System	Planned
FR-14	Estimated track is finalized in SQLite when tracking is stopped	High	System	Planned

3.3 Map & Visualization (In-flight)

ID	Description	Priority	Source	Status

FR-15	Animated position dot moves along route on offline map	**High**	User	Planned
FR-16	SIGMET zones are rendered as overlays	**High**	System	Planned
FR-17	Landmark pins appear along the route	**Medium**	System	Planned
FR-18	Altitude and speed graph shown at bottom of screen	**Medium**	User	Planned
FR-19	Current flight phase (climb/cruise/descent) indicated in UI	**Low**	User	Planned

3.4 Notifications

ID	Description	Priority	Source	Status
FR-20	Notification when position enters a SIGMET zone	**High**	System	Planned
FR-21	Notification when crossing a landmark geofence	**High**	System	Planned
FR-22	Notifications work in background	**High**	System	Planned

3.5 Post-flight Sync

ID	Description	Priority	Source	Status
FR-23	App prompts user to sync actual flight data after tracking ends and internet is available	**High**	System	Planned
FR-24	PostFlightSync fetches the actual ADS-B track from OpenSky using callsign and departure date	**High**	System	Planned
FR-25	If ADS-B data is not yet available (< 1hr after landing), app schedules a retry and notifies user when ready	**Medium**	System	Planned
FR-26	Raw ADS-B positions are normalized to match the estimated track schema (lat, lng, alt, speed, timestamp)	**High**	System	Planned
FR-27	Normalized actual track is persisted as ActualTrackPoints in SQLite	**High**	System	Planned
FR-28	User can manually trigger a re-sync if the first attempt returned incomplete data	**Low**	User	Planned
FR-29	App indicates data coverage quality (e.g., sparse oceanic vs. dense continental)	**Low**	System	Planned

3.6 Flight Comparison & Accuracy Report

ID	Description	Priority	Source	Status
FR-30	FlightComparator aligns estimated and actual tracks by timestamp using linear interpolation	**High**	System	Planned
FR-31	For each aligned pair, Haversine deviation distance is computed	**High**	System	Planned
FR-32	Max deviation, average deviation, and RMSE are calculated across the full track	**High**	System	Planned
FR-33	Phase-level accuracy is computed separately for climb, cruise, and descent	**Medium**	System	Planned
FR-34	An overall accuracy score (0–100) is generated from RMSE and max deviation	**Medium**	System	Planned
FR-35	DeviationReport is persisted to SQLite	**High**	System	Planned
FR-36	Comparison view renders both tracks overlaid on the map with color-coded deviation	**High**	User	Planned
FR-37	Deviation heat coloring: green < 10km, amber 10–50km, red > 50km	**Medium**	User	Planned
FR-38	User can scrub a timeline slider to see position at any point during the flight on both tracks	**Medium**	User	Planned
FR-39	Accuracy report card shows score, RMSE, max deviation, and per-phase breakdown	**High**	User	Planned
FR-40	User can share the accuracy report as an image or PDF	**Low**	User	Planned

3.7 Flight History

ID	Description	Priority	Source	Status
FR-41	Completed flights with comparison data are saved to history	**Medium**	System	Planned
FR-42	History shows accuracy score badge per flight	**Medium**	System	Planned
FR-43	User can replay estimated track, actual track, or both from history	**Low**	User	Planned
FR-44	User can delete flights from history	**Low**	User	Planned

4. Non-Functional Requirements

4.1 Performance

- Position update cycle must complete in under 100ms
- FlightComparator must process a full 12-hour track (1,440 points at 30s intervals) in under 3 seconds
- Post-flight sync for a typical route must complete in under 15 seconds on a 4G connection
- Deviation map must render at 60fps with up to 1,440 point-pairs overlaid

4.2 Reliability

- Dead reckoning must function with zero network calls after tracking starts
- Estimated track must survive app backgrounding and crash — session persisted to SQLite every cycle
- Post-flight sync must be idempotent — re-running does not duplicate ActualTrackPoints
- FlightComparator must handle sparse actual tracks gracefully — interpolate gaps up to 5 minutes

4.3 Accuracy

- Dead reckoning position error must remain within 50km for flights under 6 hours without manual correction
- Timestamp alignment between estimated and actual tracks must be within 60 seconds
- Accuracy score must correlate with actual deviation — a score above 80 should correspond to RMSE below 20km

4.4 Privacy

- No user data is sent to any server — post-flight sync fetches public ADS-B data only
- All data stored on-device in SQLite
- No account or login required

5. External Interface Requirements

API	Phase	Data Provided	Format	Auth
AviationAPI	**Pre-flight**	Filed routes, waypoints	JSON/REST	Free/Key
OpenSky Network	**Pre-flight**	Historical speed/altitude profiles	JSON/REST	Free
aviationweather.gov	**Pre-flight**	SIGMETs, turbulence advisories	GeoJSON	None
OurAirports	**Pre-flight**	Landmark, airspace data	CSV (static)	None
Mapbox	**Pre-flight**	Offline map tile download	Vector tiles	API Key
OpenSky Network	**Post-flight**	Actual ADS-B track by callsign + date	JSON/REST	Free
ADS-B Exchange	**Post-flight**	Fallback actual track source	JSON/REST	Free

6. System Models

The following diagrams accompany this SRS:

- System Architecture — DataPipeline, DRE, PostFlightSync, FlightComparator, SQLite, UI
- Use Case Diagram — pre-flight, in-flight, post-flight sync, comparison, history
- Sequence Diagram — in-flight track recording loop, post-flight sync, comparison computation
- ER Diagram — Flight, Route, Waypoint, SpeedProfile, Sigmet, TrackingSession, EstimatedTrackPoint, ActualTrackPoint, DeviationReport, Landmark, Notification
- UML Class Diagram — DeadReckoningEngine, DataPipeline, PostFlightSync, FlightComparator, SQLiteRepository, NotificationService, MapViewModel
- Data Flow Diagram — pre-flight and post-flight external data flows into SQLite and through comparison engine to UI

7. Appendix — Tech Stack

Layer	Technology
Framework	React Native (Expo)
Language	TypeScript
Offline DB	SQLite via expo-sqlite
Maps	react-native-maps + Mapbox Offline SDK
Background Tasks	expo-task-manager
Notifications	expo-notifications
State Management	Zustand
HTTP Client	axios (pre-flight & post-flight only)
Pre-flight Sources	AviationAPI, OpenSky, aviationweather.gov, OurAirports, Mapbox
Post-flight Sources	OpenSky Network REST API, ADS-B Exchange (fallback)
Comparison Engine	Custom TypeScript — Haversine, linear interpolation, RMSE